

James Kevius Tribble

jjtribb05@gmail.com | [linkedin.com/in/keviustribble](https://www.linkedin.com/in/keviustribble) | keviustribble.dev

EDUCATION

Clemson University

B.S. Computer Science, Minors in Mathematics and Artificial Intelligence

M.S. Computer Science, Bachelor-Graduate Program

GPA: 3.95/4.0

Clemson, SC

May 2027

May 2028

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, SQL, JavaScript

Robotics & Vision: ROS 2, PX4, Gazebo, LiDAR, CAD

ML, Data & Cloud: PyTorch, LangChain, Snowflake, PostgreSQL, AWS

Tools & Platforms: Git, GitHub, Docker, Podman, Distrobox, Kubernetes, Linux, Cursor

Certifications: AWS Certified Cloud Practitioner (In-Progress)

Relevant Coursework: Differential Equations, Linear Algebra, Theory of Probability, Computer Vision, Machine Learning, Artificial Intelligence

EXPERIENCE

Honors Thesis Researcher - Aerial Robotics

2026 - Present

Clemson University

Clemson, SC

- Developing a gripperless aerial modular assembly system in which a micro quadrotor uses passive hook, guide, ramp, and anti-backsliding geometry to pick up and place lightweight structural modules
- Built and documented a reproducible Linux robotics environment with Podman/Distrobox, Ubuntu 24.04, ROS 2 Jazzy, Gazebo Harmonic, PX4 SITL, Micro XRCE-DDS, and QGroundControl
- Modeled candidate interlocking puzzle-block modules in CAD to evaluate pickup features, guided placement, self-alignment, and tolerance to position and yaw error

Data Engineer Intern - Data Distribution & Data Share

June 2026 - August 2026

LPL Financial

Fort Mill, SC

- Analyzed and reconciled commission and mutual fund data across legacy Oracle systems and Snowflake System of Record tables, using SQL joins to verify record parity and document discrepancies
- Updated legacy SQL logic for SOR-aligned resources and mapped source fields to target schemas, accelerating validation-query development for ICAP and Direct Business Mutual Fund migrations
- Supported a PostgreSQL metadata migration by provisioning Snowflake tables and developing AWS Glue ETL pipelines to synchronize and validate critical reference data
- Applied AI-assisted development with Cursor to generate SQL validation patterns, accelerate AWS Glue implementation, and onboard to an existing enterprise data-pipeline repository

MemoryScapes - Computer Vision Engineer

August 2025 - Dec 2025

Clemson University, Senior Capstone

Clemson, SC

- Deployed containerized Nightshade research code on AWS EC2 and automated batch image-poisoning workflows with Python and Bash
- Evaluated Nightshade by training and comparing 6 Stable Diffusion XL LoRA models on clean and protected user-image datasets

Rural Robotics REU - Undergraduate Research Assistant

May 2025 - July 2025

West Virginia University, Interactive Robotics Laboratory

Morgantown, WV

- Built an iPhone-based RGB and 6-DoF pose collection workflow and automated reconstruction-data preparation for outdoor rural robotics experiments
- Benchmarked neural rendering and splatting methods using PSNR, SSIM, and LPIPS, comparing sensor-derived and structure-from-motion camera poses

NIMBUS Lab REU - Undergraduate Research Assistant

June 2024 - August 2024

University of Nebraska Lincoln

Lincoln, NE

- Developed a reinforcement-learning UAV simulation from traffic trajectory data and compared DDPG, TD3, and PPO policies for latency-aware 3D positioning

PROJECTS

Agentic Search Engine | Rust, AWS Fargate, Docker, Lambda, API Gateway, SQS, LangGraph

Summer 2026

- Engineered a search engine from scratch in Rust, building custom indexing and retrieval components to power fast, structured search across the web
- Deployed the system end-to-end on AWS using Docker, Fargate, Lambda, API Gateway, and SQS, designing cloud infrastructure to support asynchronous ingestion, scalable query processing, and reliable search result delivery
- Added an agentic search layer with LangGraph to emulate Perplexity-style research assistance, orchestrating retrieval, multi-step reasoning, and LLM-generated responses over indexed academic sources